REDIAT TEREFE

Ph.D. research scholarship

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PROFILE

Ph.D research scholar in chemical engineering at IIT Roorkee, specializing in heterogeneous catalysis for biomass valorization and intermediate upgradation. Skilled in the complete catalyst lifecycle from design and synthesis to advanced characterization and reaction parameter optimization. I have two first author journal publications on biomass conversion: one on furfural production from biomass and another on the hydrogenation of furfural-to-furfural alcohol. Additionally, I have contributed to collaborative research on tri-reforming of methane, as well as CO_2 to formic acid conversion. Looking ahead, I am keen to explore energy efficient approaches for methane and CO_2 valorization, particularly through inductive heating strategies. I am also interested in waste conversion, with a specific focus on developing sustainable pathways to convert dimethylformamide to ammonia.

EDUCATION

Ph.D., Chemical Engineering *IIT Roorkee, India | Jul 2022 – Present*

- Ph.D. title: “Production of furan intermediates from biomass waste and their upgrading to valuable chemicals using heterogeneous catalysts”.

  Supervisor: Prof. Prakash Biswas, Department of Chemical Engineering, IIT Roorkee.

M.Sc., Chemical Engineering *Jimma University, Ethiopia | Graduated Feb 2021| CGPA 3.9/4*

- Thesis: “Removal of Excess Free Fatty Acid from Edible Oil Using Betaine-Based Natural Deep Eutectic Solvents and Oil's Effect Analysis.”

  Supervisor: Prof. Ramachandran K., Department of Petroleum Engineering, JCT Institutions.

B.Sc., Chemical Engineering *Jimma University, Ethiopia | Graduated Jun 2018 | CGPA 3.4/4*

- Final-year project: “Bitumen Modification Using Waste Plastic (PET) for Road Bitumen Improvement and Minimization of Environmental Pollution.”

  Supervisor: Prof. Muluken Eshetu, Department of Chemical Engineering, JIT, Jimma University.

RESEARCH & TECHNICAL SKILLS

- Catalysis & Synthesis: Catalyst design and synthesis (MOF-derived, magnetic/separable catalysts); experiment design; process and reaction-parameter optimization (RSM).
- Characterization & Analytical Instrumentation: FE-SEM, XRD, GC, GC-MS, HPLC, FTIR, TPR/TPD/TPO, pulse chemisorption, NMR, BET.
- Computational & Programming: Python, JavaScript, MERN stack development; growing focus on AI/ML/Agentic AI applications in chemical engineering.
- Software: Origin, Design Expert, IBM SPSS, Edraw Max, X'Pert HighScore.
- Languages: English (professional working proficiency), Amharic (native), Gurage (native), Oromo (basic), Hindi (basic).

PUBLICATIONS

- Rediat T. and Prakash B. (2025). “The direct catalytic conversion of teff straw into fuel intermediate: Optimization of reaction parameters.” Biomass and Bioenergy, https://doi.org/10.1016/j.biombioe.2025.108150
- Sharma A. and Rediat T. (2025). “Optimization of reaction parameters by response surface methodology for the tri-reforming process over a Ni-silica catalyst to produce synthesis gas.” International Journal of Hydrogen Energy, https://doi.org/10.1016/j.ijhydene.2025.150378.
- Sharma A., Rediat T., Singh M., Tomer R., Steenhaut T., Tonelli A., Hermans S., Biswas P. (2026). “Amine-functionalized metal-organic framework-templated nitrogen-doped Ni-alumina catalyst for efficient methane tri-reforming.” Applied Materials Today, https://doi.org/10.1016/j.apmt.2026.103292.

- Rediat T., Sharma A., Disale S.S., Tomer R., Hermans S., Biswas P. (2026). “The conversion of furfural to furfuryl alcohol with high yield via catalytic hydrogenation transfer in the presence of nickel-embedded alumina catalyst derived from metal-organic framework template.” Catalysis Today, https://doi.org/10.1016/j.cattod.2026.115883

• CONFERENCES & ACADEMIC TRAVEL

- EuropaCat 2025, 16th European Congress on Catalysis Trondheim, Norway.
- ICECEES-2024, International chemical engineering conference on Energy, Environment and sustainability, Roorkee, India.

HONORS & AWARDS

- Certificate of Merit, Department of Chemical Engineering, Jimma Institute of Technology 1st place, final research competition (Jun 2018).
- Certificate of Recognition, Jimma University Community-Based Education Best Student Research Project, Annual Student Research Symposium (Jun 2018).
- Best research scholar’s day presentation award, annual student research presentation program (Feb 2026)

REFERENCES

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